



From Primordial Stars to Early Galaxies: A Semianalytic Model Calibrated with AEOS and Renaissance

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Abstract

We present an extension of our semianalytic model that follows the formation of Population III (Pop III) stars and their metal-enriched descendants, incorporating dark matter halo merger trees from cosmological N -body simulations and feedback from reionization. Our extended model is calibrated using two complementary cosmological hydrodynamical simulations: AEOS, which resolves individual Pop III and Population II (Pop II) stars to $z \sim 14.6$, and Renaissance, which is lower resolution but follows large-scale metal-enriched star formation to $z \sim 11$. With a combined calibration, we capture small-scale physics of primordial star formation over a large range in halo mass. We find good agreement between our calibrated model and AEOS, reproducing the evolution in number of star-forming halos and total stellar mass. Achieving this agreement requires increasing the normalization of, flattening the redshift dependence of, and adding scatter to the commonly used critical mass threshold M_{crit} . Our treatment of the delay between Pop III stellar death and subsequent Pop II star formation emphasizes the need to account for halos that have yet to transition to Pop II, since incomplete sampling of this delay in simulations limits physically motivated calibrations. Finally, we apply our model to larger-volume dark-matter-only simulations and predict that ~ 10 active Pop III sources at $z = 10$ lie within the area strongly lensed by galaxy cluster MACS J0416 with a magnification exceeding $\mu > 30$. These results demonstrate that semianalytic approaches, when calibrated to hydrodynamical simulations, can provide accurate, computationally efficient predictions for the earliest stages of cosmic star formation.

Unified Astronomy Thesaurus concepts: Population III stars (1285); Galaxy formation (595); High-redshift galaxies (734); Cosmology (343)

1. Introduction

The direct descendants of the first stars, the most metal-poor stars that we observe in our own Milky Way halo and in the Local Group, currently provide the best observational insight into the properties of the first metal-free stars, designated Population III (Pop III), and the first galaxies that formed at redshifts $30 < z < 20$ (A. Frebel & J. E. Norris 2015; R. S. Klessen & S. C. O. Glover 2023). Although they existed only briefly in each galaxy's evolution, Pop III stars marked the onset of cosmic chemical evolution by forming the first elements heavier than Li. These newly forged metals mixed into the surrounding gas, setting the initial abundance patterns inherited by the next generation of metal-poor Population II (Pop II) stars.

Direct observations of Pop III stars remain elusive owing to their short lifetimes and formation predominately at high redshift. However, recent results suggest that their signatures may now be within reach. Rare cases of strong gravitational lensing can amplify faint stellar clusters to observable levels

(R. A. Windhorst et al. 2018; A. T. P. Schauer et al. 2020; E. Vanzella et al. 2023a; A. Adamo et al. 2024; L. Mowla et al. 2024). The lensed system LAP1-B at $z = 6.6$ shows multiple hallmarks of Pop III star clusters: formation in pristine halos, and a top-heavy initial mass function (IMF; E. Visbal et al. 2025; K. Nakajima et al. 2026). Several high-redshift galaxies exhibit strong He II $\lambda 1640$ emission or very low metallicities (E. Vanzella et al. 2023b; R. Maiolino et al. 2024; X. Wang et al. 2024; S. Cai et al. 2025; S. Fujimoto et al. 2025; T. Morishita et al. 2025), consistent with signatures related to Pop III stars.

There are also a range of complementary observables that provide indirect constraints on the first stars. These include transient phenomena that probe Pop III remnants, including supernovae (SNe; T. Hartwig et al. 2018; T. J. Morioka et al. 2019) and gamma-ray bursts (K. Toma et al. 2016), and gravitational waves from black hole mergers (K. Inayoshi et al. 2016; M. Safarzadeh & Z. Haiman 2020; B. Liu et al. 2024). Absorption studies in objects like the recently identified damped Ly α systems (DLAs) at $z \sim 6$ show abundance ratios, particularly [C/O] and [Si/O], that point to enrichment from Pop III SNe (A. Sodini et al. 2024; E. Visbal et al. 2026). Global measurements reflecting Pop III star formation include the optical depth of the cosmic microwave background (E. Visbal et al. 2015b), the cosmological 21 cm signal



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(J. Mirocha et al. 2018; T. Gessey-Jones et al. 2022), and He II $\lambda 1640$ line intensity mapping (E. Visbal et al. 2015a; J. Parsons et al. 2022). Finally, stellar archeology can reveal how Pop III SN ejecta are imprinted on the next generation of metal-poor stars (A. Frebel & J. E. Norris 2015). Interpreting these diverse observational probes requires a theoretical framework for the properties of the first stars. Of particular interest is constraining the uncertainty in the Pop III IMF, as well as their lifetimes, feedback, and chemical enrichment, which imprint their signatures on subsequent generations of stars and galaxies.

The Pop III IMF is thought to be top-heavy, due to the slow cooling of primordial clouds through the H_2 rovibrational line (Z. Haiman et al. 1996; M. Tegmark et al. 1997), which would result in large Jeans masses and stellar masses on the order of 10^2 – $10^3 M_\odot$ (T. Abel et al. 2002; V. Bromm et al. 2002, 2004; N. Yoshida et al. 2008; S. Hirano et al. 2015; T. Hosokawa et al. 2016). Other recent theoretical work has demonstrated that Pop III stars may actually form with characteristic masses of tens of solar masses or lower (A. Stacy et al. 2010; P. C. Clark et al. 2011; T. H. Greif et al. 2011, 2012; T. Hosokawa et al. 2011, 2016; M. A. Latif et al. 2013, 2022; S. Hirano et al. 2014; P. Sharda et al. 2020), or that they may even form with masses below $0.8 M_\odot$ (O. Jaura et al. 2022; W. Lake et al. 2025; P. Sharda & S. H. Menon 2025). An IMF with a power-law slope shallower than E. E. Salpeter (1955) is supported by observations of metal-poor dwarf galaxies (M. Geha et al. 2013), which support the hypothesis of an evolving IMF that was bottom-light at early times. A flat IMF for massive stars (10 – $300 M_\odot$) is disfavored by observations of carbon-enhanced metal-poor stars (M. de Bannassuti et al. 2017). Additionally, a few individual metal-poor stars have been observed with abundances that suggest that they have been enriched by a single SN (R. Ezzeddine et al. 2019; Á. Skúladóttir et al. 2021, 2024; D. Yong et al. 2021; Q.-F. Xing et al. 2023; A. P. Ji et al. 2024) and could be used to provide constraints on the yields of Pop III stars. However, the interpretation of these abundances and their connection to chemical yields is sensitive to the retention and mixing of ejected metals into the interstellar medium (ISM; see, e.g., J. Mead et al. 2025a, 2025b).

Understanding the observations of extremely metal-poor stars and their implications for the properties of Pop III stars and galaxies requires computational models of galaxy formation and evolution complete from the first stars to the present day. One avenue to pursue is through numerical hydrodynamic simulations that follow the evolution of dark matter (DM) and star particles, as well as gas. In particular, studying the formation, evolution, and effects of Pop III stars requires high spatial and mass resolution in simulations that model individual stars (A. Emerick et al. 2019; N. Lahén et al. 2020, 2023; T. A. Gutcke et al. 2021; E. P. Andersson et al. 2023, 2025; K. Brauer et al. 2025a) in order to capture sites of Pop III star formation in minihalos ($10^5 M_\odot < M_h < 10^6 M_\odot$) and the impact of individual stellar feedback on the chemodynamical evolution of their environment. However, the high computational costs of modeling individual stars, tracking yields for multiple heavy elements, and tracing radiative feedback, all at high resolution, make developing a cosmological model to study the effects of Pop III stars on present-day observations difficult. High computational costs also prevent thorough exploration of parameter space for

assumed values and functions such as the IMF, SN injection energy, DM streaming velocities, etc., all of which impact the contribution of Pop III to the chemodynamical evolution of galaxies.

Semianalytic models (SAMs) are computationally efficient alternatives to full hydrodynamic cosmological simulations, running orders of magnitude faster. By combining DM halo merger trees with analytic prescriptions for star formation, radiation, feedback, metal enrichment, and so forth, SAMs can be used to rapidly explore parameter space to study the effects of altering one or more of the many tunable parameters in the models. Importantly, SAMs also enable studies on the larger domain sizes needed to statistically capture the stochastic variation in star formation, metal enrichment, and reionization expected in the early Universe. Although many SAMs do not implement modeling for the first stars, instead initializing a metallicity floor representative of their enrichment, there are now numerous SAMs that have been used to study Pop III and the first galaxies (e.g., M. Trenti & M. Stiavelli 2009; Y. Komiya et al. 2010; B. D. Crosby et al. 2013; B. Liu & V. Bromm 2020; E. Visbal et al. 2020; L. C. Corazza et al. 2022; T. Hartwig et al. 2022; S. Hegde & S. R. Furlanetto 2023; C. R. Feathers et al. 2024; E. M. Ventura et al. 2024; T. Ishiyama & S. Hirano 2025). Ultimately, SAMs with Pop III models, run and calibrated to present-day observations, are critical to the exploration of broad parameter spaces to interpret the observational signatures from Pop III and their descendants.

Calibrating a SAM for Pop III star formation and the transition to Pop II requires simulations that resolve both the first Pop III-forming halos and those that subsequently host Pop II. Because only a few Pop III stars form per halo, they must be modeled as individual stars drawn from an IMF rather than as stellar population particles; otherwise, stellar masses converge toward the particle mass resolution (K. Brauer et al. 2025b). Additionally, the feedback from individual stars is weaker and less bursty than that from particles containing single stellar populations (M. C. Smith 2021), whose use can yield higher stellar masses at a fixed halo mass (K. Brauer et al. 2025b; M. Jeon & M. Ko 2026). Thus, calibrating a SAM to simulations that track individual Pop III stars is essential to capture the evolution of both primordial and early metal-enriched environments.

In this paper, we calibrate a SAM (E. Visbal et al. 2020) to the AEOS simulations (K. Brauer et al. 2025a) in order to develop a computationally efficient model of Pop III star formation and galaxy evolution in minihalos. We summarize the relevant details from the AEOS simulations and the E. Visbal et al. (2020) SAM in Section 2. Our method for calibrating the SAM to AEOS is described in Section 3, with an analysis of the resulting Pop III and Pop II star formation in the SAM in Section 4. We then apply the calibrated SAM to a selection of other cosmological volumes and compare the predicted number density of active Pop III sources to observations in Section 5, and we conclude in Section 6.

2. Computational Methods

We summarize here the relevant physics and model components used in the AEOS simulations (K. Brauer et al. 2025a) and the Renaissance-calibrated SAM (E. Visbal et al. 2020; R. Hazlett et al. 2025), and we refer the reader to the respective methods papers for greater detail.

2.1. AEOS

AEOS (K. Brauer et al. 2025a) is a 1 comoving Mpc per side cosmological hydrodynamics simulation run with the adaptive mesh code ENZO (G. L. Bryan et al. 2014; C. Brummel-Smith et al. 2019) that uses the best-fit cosmological parameters from Planck Collaboration et al. (2014). Star particles in AEOS represent individual massive stars that contribute feedback on the timescale of the simulation, enabling the study of the effects of individual stellar feedback. With a DM mass resolution of $1840 M_\odot$ and a maximum spatial resolution of 1 proper pc, AEOS can resolve minihalos ($M_{\text{vir}} \sim 10^5 M_\odot$), which are thought to be the formation sites of the first Pop III stars. We limit our summary of AEOS here to the details relevant for the calibration of the SAM.

2.1.1. Radiative Cooling

AEOS uses a modified version of GRACKLE (B. D. Smith et al. 2017) to follow radiative heating and cooling from a F. Haardt & P. Madau (2012) UV background that is extended to high redshifts and metal lines. Additionally, AEOS includes background Lyman–Werner (LW) contributions adopted from A. Emerick et al. (2019) and J. H. Wise et al. (2012), with a high-redshift contribution adopted from Y. Qin et al. (2020).

2.1.2. Star Formation

Stars in the simulation form stochastically from an IMF when at least $100 M_\odot$ of gas in a given cell satisfy the following star formation criteria: (i) converging gas flow ($\nabla \cdot \mathbf{v} < 0$), (ii) $T < 500$ K, and (iii) $n > 10^4 \text{ cm}^{-3}$.

Population III. In addition to the above constraints, gas that has a minimum molecular hydrogen fraction $f_{\text{H}_2} > 0.0005$ (H. Susa et al. 2014) and metallicity $Z < 10^{-5} Z_\odot$ goes on to form Pop III stars. Following J. H. Wise et al. (2012), Pop III stellar masses are drawn from an E. E. Salpeter (1955) IMF above a characteristic mass M_{char} and an exponential cutoff below. Lifetimes for these stars are adopted from D. Schaerer (2002). For our calibration, we use the AEOS10 simulation, which uses $M_{\text{char}} = 10 M_\odot$ and a mass range of $1 M_\odot < M_* < 100 M_\odot$.

Population II. Pop II stellar masses are drawn from a P. Kroupa (2001) IMF with a range of $0.08\text{--}120 M_\odot$. Pop II stars are treated as individual particles if their mass exceeds $2 M_\odot$. The remaining stars are agglomerated into single particles, as they provide neither feedback nor chemical enrichment during the simulation. Lifetimes and asymptotic giant branch (AGB) phases are set using PARSEC (A. Bressan et al. 2012).

2.1.3. Stellar Feedback

AEOS implements both AGB and massive stellar winds, core-collapse SNe, Type Ia SNe, and ionizing radiation from massive stars. Mass-loss rates from winds are assumed to be constant over the relevant ejection phase. All SNe explode with a fixed energy of 10^{51} erg, with core-collapse SNe occurring for Pop III stars with $10 M_\odot < M_* < 100 M_\odot$ and Pop II stars with $8 M_\odot < M_* < 25 M_\odot$, and Type Ia SNe occurring for Pop II stars with $3 M_\odot < M_* < 8 M_\odot$. Ionizing radiation from Pop III stars uses luminosities from A. Heger & S. E. Woosley (2010) and lifetimes from D. Schaerer (2002), while Pop II luminosities are taken from OSTAR2002

(T. Lanz & I. Hubeny 2003) when available; otherwise, they are taken from a scaled blackbody spectrum.

2.2. Semianalytic Model Calibrated to Renaissance

We use an updated version of the SAM presented in E. Visbal et al. (2020), which takes DM halo merger trees with 3D spatial coordinates from cosmological N -body simulations and applies analytic prescriptions for Pop III and metal-enriched star formation. The model is calibrated to the Renaissance simulations (B. W. O’Shea et al. 2015), cosmological hydrodynamics simulations performed with the adaptive mesh refinement code ENZO (G. L. Bryan et al. 2014; C. Brummel-Smith et al. 2019) that model the formation of the first stars and galaxies. The SAM also includes feedback from LW radiation and the expansion of H II regions, computed on a grid using fast Fourier transforms, along with external metal enrichment.

Whether a halo forms Pop III stars or not is a key parameter in the model. With a DM particle mass of $2.9 \times 10^4 M_\odot$, halos with $M_h \lesssim 2 \times 10^6 M_\odot$, including $10^5\text{--}10^6 M_\odot$ minihalos where Pop III stars are thought to form in the early Universe, are poorly resolved (B. W. O’Shea et al. 2015). This limits the robustness of the Pop III calibration using Renaissance. In contrast, AEOS resolves a large sample of such minihalos, allowing a more reliable calibration that can be applied over a broader range of halo mass.

The Renaissance-calibrated SAM described by R. Hazlett et al. (2025) reproduces both Pop III and metal-enriched star formation rate densities (SFRDs) within a factor of ~ 2 compared to the calibrating simulation between $11 < z < 25$ in a representative cosmological volume. For regions that are more overdense or rarefied, we found metal-enriched SFRDs that agreed with Renaissance within a factor of ~ 2 at high redshift but differ by a factor of ~ 3 or more at later times.

3. Calibration to AEOS

To calibrate the SAM to AEOS, we identify halos that host the earliest Pop III star formation in AEOS and track their subsequent metal-enriched star formation. We determine the critical halo mass for Pop III formation and the stellar masses of Pop III stars, introducing a key modification to M_{crit} that better reproduces the distribution of halos that form Pop III stars in AEOS. We measure the delay times before the onset of Pop II star formation and parameterize metal-enriched Pop II starbursts in low-mass halos using power-law fits, combining these with the Renaissance-calibrated bursts for higher-mass halos. Finally, we implement steady star formation and feedback processes, including H II bubble growth and external metal enrichment, to produce a self-consistent model that reproduces the timing, mass, and distribution of Pop III and Pop II star formation observed in AEOS.

3.1. Halo Sample

We identify DM halos using the halo finder ROCKSTAR (P. S. Behroozi et al. 2013a) and calculate halo merger trees with CONSISTENT TREES (P. S. Behroozi et al. 2013b). For our calibration we focus on the 216 halos in AEOS that host the first instances of Pop III star formation, along with the stellar mass histories for ~ 20 halos that experience subsequent metal-enriched star formation.

3.2. Star Formation Parameterization

Our model tracks Pop III star formation in pristine halos and the subsequent onset of metal-enriched stars that occurs after a delay period. Metal-enriched star formation then proceeds through a bursty phase of rapid starbursts separated by quiescent periods, followed by a steady phase once halos reach a critical virial temperature allowing sustained star formation. This critical temperature corresponds to the halo mass below which photoheating from reionization suppresses star formation (e.g., P. R. Shapiro et al. 1994; A. A. Thoul & D. H. Weinberg 1996; N. Y. Gnedin & L. Hui 1998; N. Y. Gnedin 2000; M. Dijkstra et al. 2004; M. Hoeft et al. 2006; T. Okamoto et al. 2008; E. Sobacchi & A. Mesinger 2013; Y. Noh & M. McQuinn 2014). Rather than modeling individual stars, the SAM calculates the total stellar mass formed in each halo over time intervals set by the DM merger trees.

It is important to note that our star formation prescription differs from the calibration presented in R. Hazlett et al. (2025) in several key ways. We calibrate Pop III star formation in our model exclusively to AEOS and exclude prior calibration to the Renaissance simulations. This is due to the improved resolution of the minihalos hosting Pop III formation in AEOS. We also calibrate to metal-enriched star formation, extending the metal-enriched burst mass relation from Renaissance to lower halo masses. We discuss details for our star formation prescription calibrated to AEOS in the following section.

3.2.1. Population III

Pop III stars form when pristine gas collapses to high densities, enabled by H_2 and atomic cooling. This is parameterized by M_{crit} , the critical halo mass for Pop III star formation to occur, which strongly influences early star formation rates in the SAM. The value of M_{crit} depends on several processes. LW radiation dissociates H_2 , raising M_{crit} in regions with strong LW backgrounds (A. Dekel & M. J. Rees 1987; Z. Haiman et al. 1997, 2000; M. E. Machacek et al. 2001; J. H. Wise & T. Abel 2007; B. W. O’Shea & M. L. Norman 2008; K. Ahn et al. 2009; E. Visbal et al. 2014), although self-shielding in minihalos can partially offset this effect. In addition, the baryon-DM streaming velocity v_{bc} suppresses gas accretion, further increasing M_{crit} in high-velocity regions (M. Kulkarni et al. 2021; A. T. P. Schauer et al. 2021; O. Nebrin et al. 2023).

For our model, we compute the critical mass using a modified version of the fitting function found in M. Kulkarni et al. (2021) that incorporates H_2 self-shielding, along with a dependence on the LW background flux J_{LW} and the baryon-DM streaming velocity:

$$M_{\text{crit}} = M_{z=20}(J_{\text{LW}}, v_{\text{bc}}) \left(\frac{1+z}{21} \right)^{-\alpha(J_{\text{LW}}, v_{\text{bc}})}, \quad (1)$$

where

$$M_{z=20}(J_{\text{LW}}, v_{\text{bc}}) = 2.25 \times (M_{z=20})_0 (1 + J_{\text{LW}}/J_0)^{\beta_1} \times (1 + v_{\text{bc}}/v_0)^{\beta_2} (1 + J_{\text{LW}} v_{\text{bc}}/J_0 v_0)^{\beta_3} \quad (2)$$

and

$$\alpha(J_{\text{LW}}, v_{\text{bc}}) = \alpha_0 (1 + J_{\text{LW}}/J_0)^{\gamma_1} (1 + v_{\text{bc}}/v_0)^{\gamma_2} \times (1 + J_{\text{LW}} v_{\text{bc}}/J_0 v_0)^{\gamma_3} - 0.8. \quad (3)$$

The constants in these functions are given in M. Kulkarni et al. (2021).

To align with the physical conditions in AEOS, we employ a J_{LW} equivalent to the LW background described in Section 2.1.1 and $v_{\text{bc}} = 0 \text{ km s}^{-1}$. To produce a similar distribution of halos that form Pop III stars, we modified the fit from M. Kulkarni et al. (2021) by multiplying $M_{z=20}$ by 2.25 and reducing α by 0.8 to increase M_{crit} and flatten the slope at high redshift (Equation (1)). Additionally, we implement a lognormal scatter to M_{crit} with a standard deviation of $\log_{10}(M_{\text{crit}}/M_{\odot}) = 0.4$. This scatter is assigned at halo formation and passed along to descendant halos. If there is a merger between halos, a random standard deviation from one of the halos is assigned to the descendant.

Figure 1 shows the original and modified functions. This modification accounts for how the scatter in M_{crit} interacts with the halo mass function in merger tree realizations. While the original fitting function of M. Kulkarni et al. (2021) defines M_{crit} as the mass where 50% of halos contain sufficient cold, dense gas, applying this definition directly leads to a bias toward lower-mass halos in our SAM. Because there are many more halos just below M_{crit} than above it, the introduction of scatter causes a systematic excess of low-mass Pop III-forming halos compared to AEOS. By increasing $M_{z=20}$ and flattening the redshift dependence, our modified M_{crit} better represents the subset of halos that actually cross the threshold for Pop III star formation in AEOS, while remaining broadly consistent with the $v_{\text{bc}} = 0$ and $0 < J_{21} < 1$ results shown in Figure 3 of M. Kulkarni et al. (2021).

Figure 1 shows the distribution of DM halo masses hosting Pop III star formation as a function of redshift. Our modifications to Equation (1) result in our SAM producing both a similar number and a similar overall distribution of Pop III-forming halos to AEOS. This modification to M_{crit} is a key result of our work, as it ensures that the SAM accurately reproduces the number and distribution of Pop III-forming halos seen in AEOS. Taking into account how the scatter in M_{crit} interacts with the halo mass function, the adjustment prevents the systematic overproduction of low-mass Pop III halos shown with the purple crosses in Figure 1. Any SAM that aims to model early star formation across a realistic halo mass range should incorporate this modified M_{crit} to reliably capture the timing and sites of Pop III star formation.

To determine the overall minimum mass for Pop III star formation, we take the minimum of M_{crit} and the atomic cooling threshold found in hydrodynamical cosmological simulations (R. Fernandez et al. 2014),

$$M_a = 5.4 \times 10^7 \left(\frac{1+z}{11} \right)^{-3/2} M_{\odot}. \quad (4)$$

We calibrate Pop III star formation in our SAM to produce stellar masses consistent with AEOS. We parameterize the stellar masses of a given star formation event in an AEOS halo using the number of Pop III star-forming cells in the halo between one time step and the next. We parameterize star formation in this way because AEOS forms stars in a given

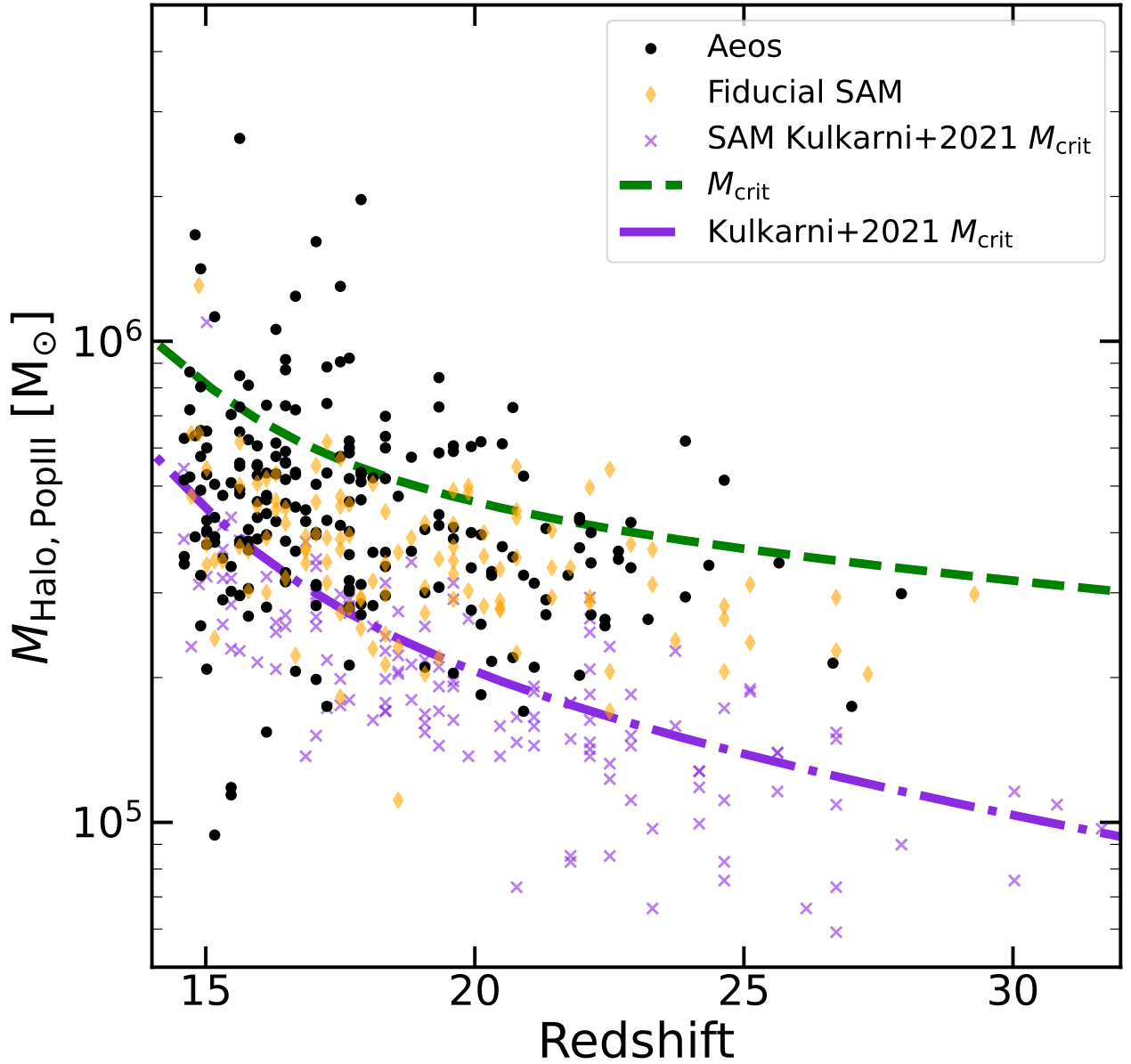


Figure 1. Virial masses for halos hosting the first Pop III stars in AEOS (black points) compared to a single realization of our SAM (orange diamonds) and a realization of the SAM using an unmodified M_{crit} from M. Kulkarni et al. (2021), shown with purple crosses. The green dashed line shows our modified M_{crit} from Equation (1), while the purple dashed-dotted line shows the unmodified M_{crit} from M. Kulkarni et al. (2021). Our modification corrects for the effect introduced by applying scatter to M_{crit} , which otherwise produces an artificial excess of low-mass Pop III-forming halos. This adjustment brings the distribution of Pop III host halos into closer agreement with AEOS.

cell once the gas mass available for star formation exceeds $100 M_{\odot}$, resulting in a buildup of masses at multiples of $100 M_{\odot}$ for a given star formation event. Thus, it is easier to define a distribution of star-forming cells than the total stellar mass for a star formation event. The cumulative distribution of the number of cells that form stars during a given event is shown in Figure 2. We fit the functional form

$$F(x) = 2.29e^{-1.46x} + 0.0643e^{-0.235x} \quad (5)$$

to this distribution, where x is the number of cells that form stars in a given Pop III star formation event in AEOS. In the SAM, to determine the mass of Pop III stars formed in a given halo after its mass exceeds M_{crit} , we probabilistically sample an x from Equation (5) and then sample stellar masses from the

same Salpeter IMF as AEOS, as described in Section 2.1.2, until the last sampled star reaches or exceeds $x \times 100 M_{\odot}$.

3.2.2. Population II

Following enrichment from Pop III stars, metals are dispersed into the nearby ISM and intergalactic medium, fueling the formation of Pop II stars. If a halo is in a bubble that exceeds our fiducial critical metallicity of $Z_{\text{crit}} = 10^{-5} Z_{\odot}$, Pop II star formation can proceed. Compared to radiative feedback, metals take comparatively longer to propagate throughout the host halo and to nearby halos. To model the time it takes for the gas to recover from feedback effects following the birth of the first Pop III stars in a halo, we introduce a delay time, t_{delay} , before the beginning of Pop II star formation.

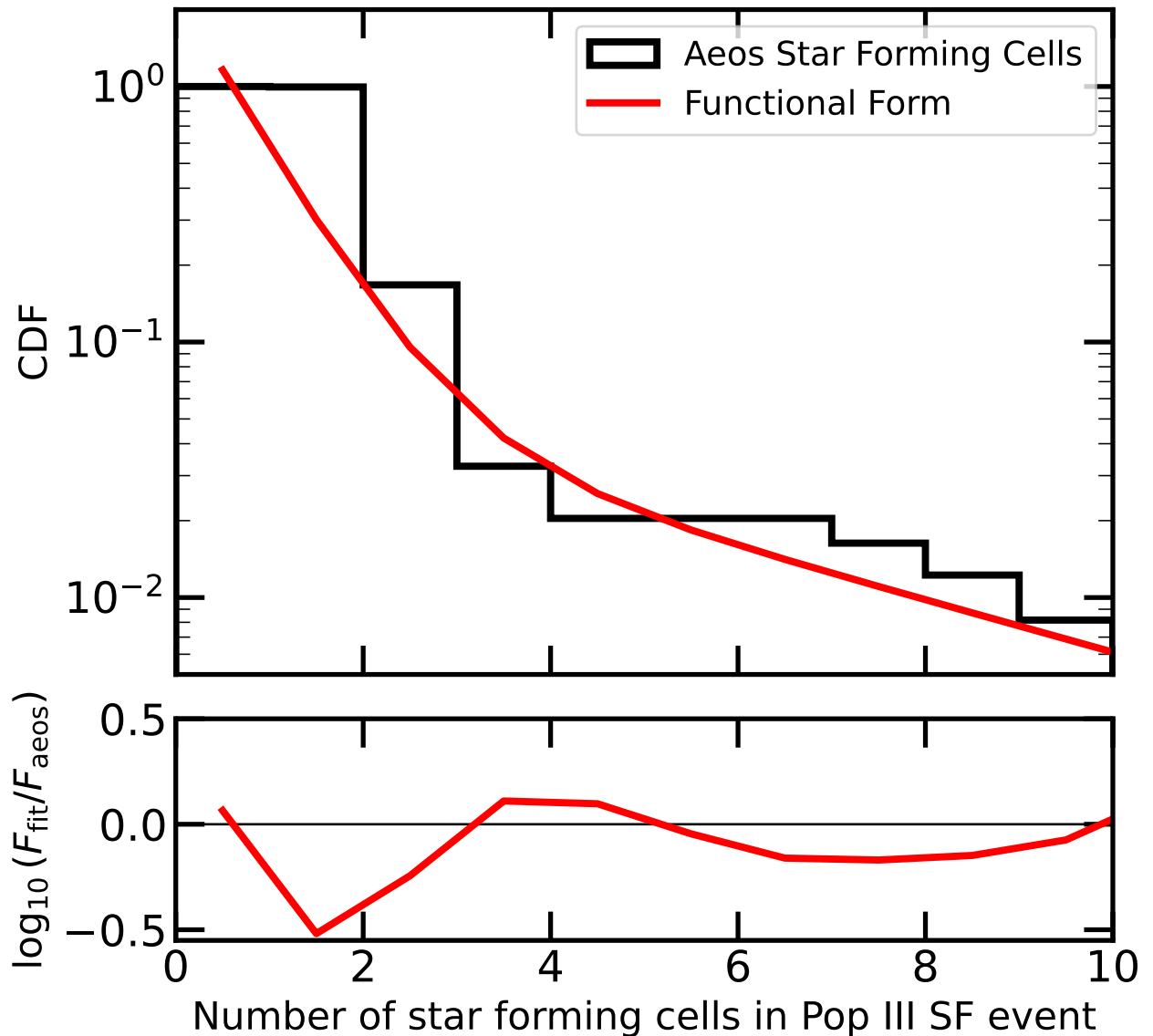


Figure 2. Cumulative distribution of the number of star-forming cells in a Pop III star formation event in AEOS. The black histogram shows the distribution measured from AEOS, where the number of star-forming cells is estimated by dividing the total stellar mass formed in each event by $100 M_{\odot}$. The red curve shows the best-fit cumulative distribution function used in the SAM (Equation (5)). The bottom panel shows the logarithmic residuals between the best-fit functional form and the AEOS cumulative distribution. This distribution is sampled by the SAM to determine the number of star-forming cells in a Pop III star formation event before assigning stellar masses from the Pop III IMF.

In Figure 3, we show the distribution of t_{delay} for AEOS halos compared to a single realization of the SAM. We find good agreement with AEOS using a lognormal distribution centered on $t_{\text{delay}} \approx 300$ Myr with a standard deviation of $\log_{10}(t_{\text{delay}}/1 \text{ Myr}) = 0.650$. Note that the adopted t_{delay} distribution is not a direct fit to the AEOS delay times. Because the final snapshot of AEOS occurs at $z = 14.6$, only halos with relatively short delay times can complete the transition from Pop III to Pop II star formation within the simulation. As a result, the AEOS sample primarily probes the short delay tail of the underlying distribution, while halos with longer t_{delay} remain unobserved. As shown in Figure 3, our lognormal distribution broadly reproduces the 16 AEOS delay times between 10 and 100 Myr but underestimates the seven AEOS halos with delays between 1 and 10 Myr. However, we find that this distribution produces approximately the same number of Pop II-forming halos as AEOS and provides a reasonable description of the overall delay time population.

We emphasize that the adopted lognormal distribution is not unique; other underlying distributions may also reproduce the AEOS sample and the number of Pop II-forming halos. The true underlying t_{delay} distribution therefore remains uncertain and cannot be fully constrained by simulations such as AEOS that conclude during the early stages of reionization.

We also tested setting t_{delay} to a constant value of 100 Myr and found that our SAM produced roughly an order of magnitude more Pop II-hosting halos than AEOS for $z < 25$, along with many orders of magnitude higher total Pop II stellar mass between $20 < z < 25$. This strong sensitivity to t_{delay} demonstrates that it is a key parameter governing the overall star formation history. The underlying physics that determines t_{delay} remains uncertain but could be related to how radiative feedback and SN feedback couple to the gas (M. Jeon et al. 2014), how efficiently metals mix within halos, and how rapidly gas can be reaccreted to form subsequent generations of stars.

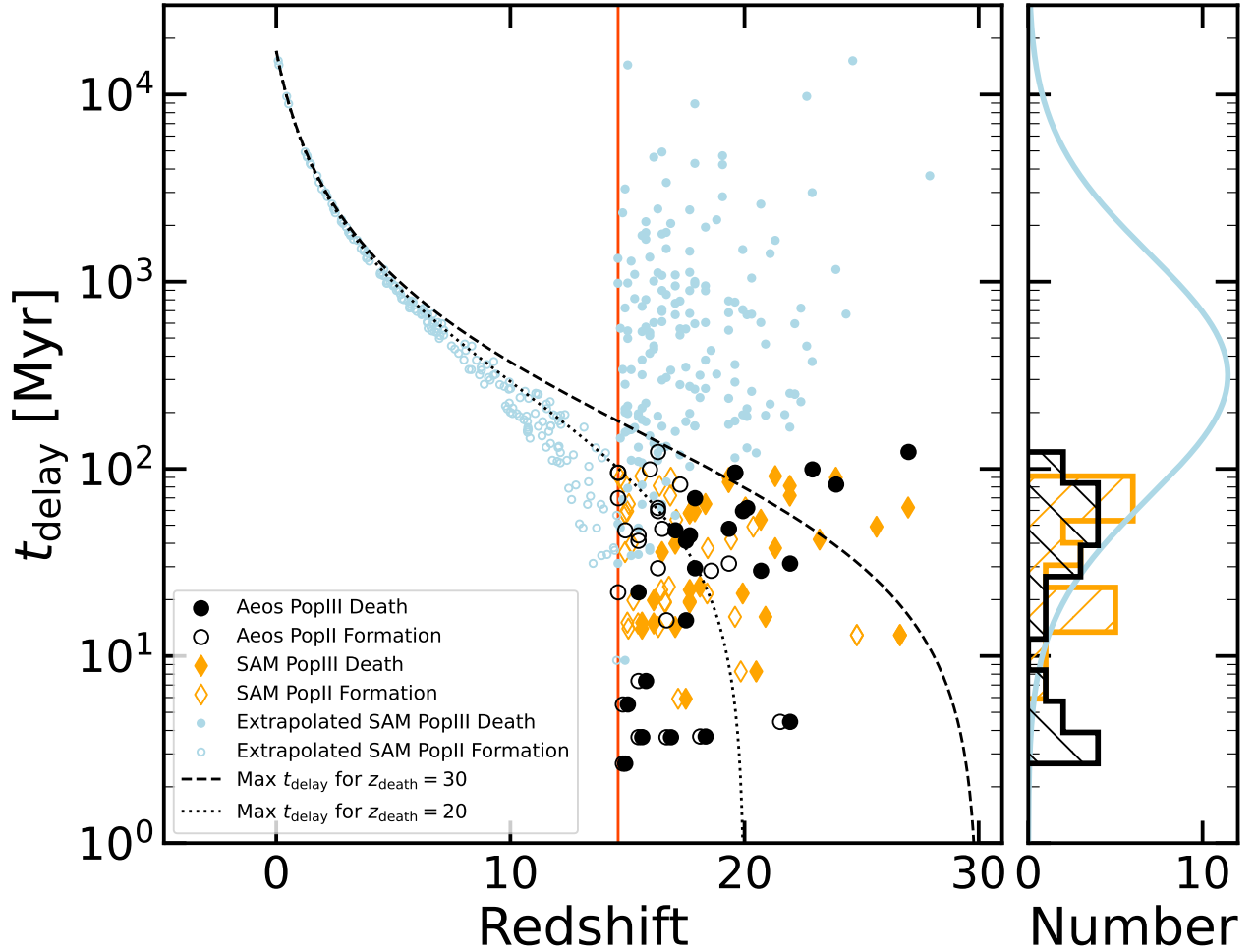


Figure 3. The delay period from the death of the first Pop III star (filled points) to the initial formation of metal-enriched Pop II stars (open points) in a single halo. The vertical red solid line marks the final redshift of AEOS at $z = 14.6$. The orange diamonds and light-blue points show the estimated t_{delay} values from a single realization of the SAM for halos with Pop II formation at $z \geq 14.6$ and extrapolated to $z < 14.6$, respectively. The dashed and dotted lines represent the maximum possible t_{delay} that can be observed for halos with Pop III deaths at $z = 30$ and $z = 20$. In the right panel, the black histogram shows the AEOS t_{delay} distribution, while the orange histogram shows the SAM t_{delay} distribution for halos forming Pop II stars before $z = 14.6$. The light-blue curve shows the adopted lognormal distribution for t_{delay} used by the SAM. The distribution is not a direct fit to the AEOS delays and includes longer delay times that are not observable before the final simulation redshift.

By the final snapshot at $z = 14.6$, most halos in AEOS that have experienced internal metal enrichment are actively forming stars in the bursty stage of Pop II star formation. In these halos, we assume that once the delay time t_{delay} has elapsed after Pop III formation, sufficient metals are present to trigger Pop II formation. In contrast, for externally enriched halos, Pop II formation depends explicitly on the metallicity of the halo exceeding the critical threshold, as described in the following subsection.

We evaluated multiparameter power-law fits to the halo virial mass and redshift at the onset of the starburst using the SCIPY curve-fit package and applied them to the AEOS distribution. We find that the Pop II stellar mass formed in a burst is correlated with the halo virial mass and increases less steeply than in the Renaissance halos, as shown in Figure 4. Unlike the Renaissance sample, we find only a weak correlation between starburst redshift and stellar mass, likely due to the small number of Pop II-forming halos in the AEOS volume, and that most of the starbursts occur at a similar redshift. We parameterize the metal-enriched stellar mass

formed in a burst as

$$M_{\text{burst,aeos}} = \alpha_{\text{aeos}} \left(\frac{M_{\text{vir}}}{10^6 M_{\odot}} \right)^{\gamma_{\text{aeos}}} M_{\odot}, \quad (6)$$

where M_{vir} is the virial mass of the halo. The best-fit parameters are $\alpha_{\text{aeos}} = 2.051 \times 10^3$ and $\gamma_{\text{aeos}} = 0.498$. We find that the distribution of the sample around $M_{\text{burst,aeos}}$ is lognormal with a standard deviation of $\log_{10}(M_{\text{burst,aeos}}/M_{\odot}) = 0.713$. As halos reach larger masses at lower redshifts, the stellar mass of the Pop II starburst increases. In general, the distribution and best fit for $M_{\text{burst,aeos}}$ are consistent with the Renaissance simulations and extend the prescription introduced in R. Hazlett et al. (2025) to lower-mass halos that could not be resolved.

To combine the Pop II star formation calibrations in AEOS and Renaissance, we implement a halo mass threshold separating the AEOS-calibrated Pop II bursts from the Renaissance-calibrated bursts in higher-mass halos. Based on the distribution of AEOS and Renaissance starbursts in Figure 4, halos in the bursty stage with masses

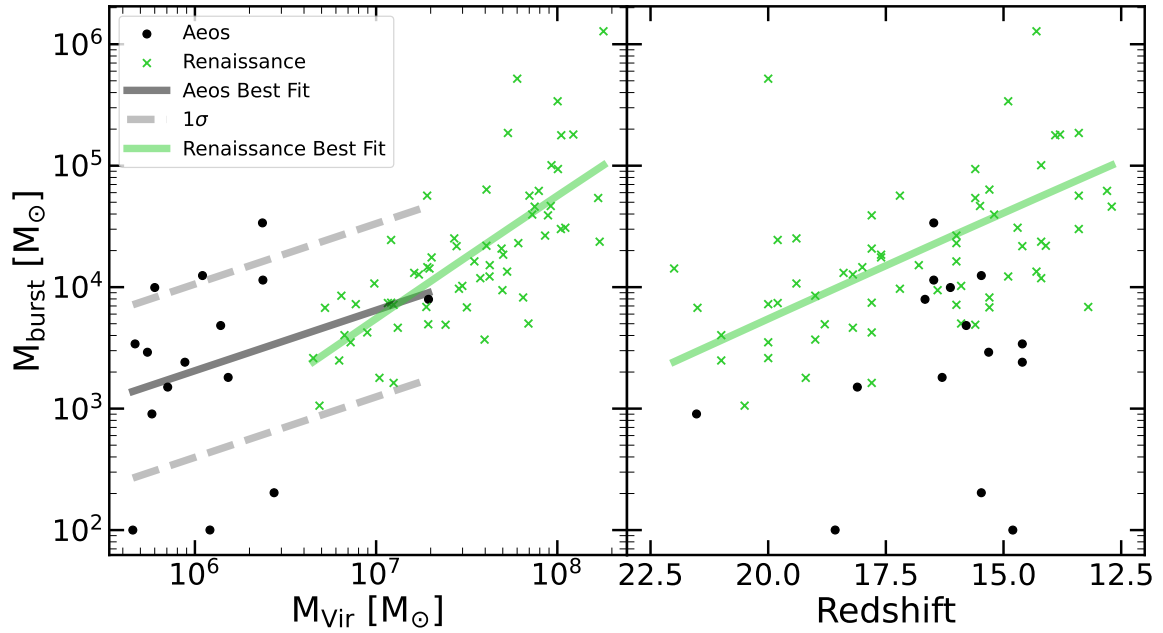


Figure 4. Metal-enriched Pop II starburst masses for the AEOS halos (black points) and the sample of Renaissance halos (green crosses) from R. Hazlett et al. (2025) during the bursty stage compared with the halo virial mass and redshift at the starburst. The best-fit power law (solid) and 1σ lines (dashed) are shown. We did not find a significant redshift dependence in the AEOS data for M_{burst} , so only the best-fit curve using halo mass is shown in the left panel. Overall, the AEOS halos extend the burst mass relation to lower-mass halos than Renaissance.

Table 1
Parameters for Star Formation in the Semianalytic Model

Parameter	Description	Fiducial Equation	Fit Values	Standard Deviation
t_{delay}	Delay before Pop II star formation	...	300 Myr	0.650
$M_{\text{burst,aeos}}$	AEOS Pop II starburst mass	$\alpha_{\text{aeos}} \left(\frac{M_{\text{vir}}}{10^9 M_{\odot}} \right)^{\gamma_{\text{aeos}}} M_{\odot}$	$\alpha_{\text{aeos}} = 2.051 \times 10^3$ $\gamma_{\text{aeos}} = 0.498$	0.713
$M_{\text{burst,ren}}$	Renaissance Pop II starburst mass	$\alpha_{\text{ren}} \left(\frac{M_{\text{vir}}}{10^9 M_{\odot}} \right)^{\gamma_{\text{ren}}} \left(\frac{1+z}{20} \right)^{\delta_{\text{ren}}} M_{\odot}$	$\alpha_{\text{ren}} = 5.36 \times 10^3$ $\gamma_{\text{ren}} = 1.092$ $\delta_{\text{ren}} = 0.578$	0.4079
t_{quies}	Quiescent period after Pop II starburst	$\epsilon \left(\frac{M_{\text{vir}}}{10^9 M_{\odot}} \right)^{\kappa} \text{ Myr}$	$\epsilon = 44.8$ $\kappa = -0.394$	0.255
f_{II}	Steady stage star formation efficiency	$\dot{M}_{*} = \frac{f_{\text{II}}}{t_{\text{ff}}} M_{\text{gas}},$ $\dot{M}_{\text{gas}} = \dot{M}_{\text{acc}} - \dot{M}_{*} - \eta_{\text{SN}} \dot{M}_{*}$	$f_{\text{II}} = 0.0129$	0.422

Note. Parameters t_{delay} , $M_{\text{burst,aeos}}$, $M_{\text{burst,ren}}$, t_{quies} , and f_{II} each follow a lognormal distribution. The last column shows the standard deviation of the $\log_{10}$ of the corresponding quantity mentioned in the text. Halos in the bursty stage with masses $< 4 \times 10^6 M_{\odot}$ will have Pop II burst masses set by $M_{\text{burst,aeos}}$, and halos with masses $\geq 4 \times 10^6 M_{\odot}$ instead use $M_{\text{burst,ren}}$. Halos with masses $> 10^9 M_{\odot}$ extrapolate beyond the calibration to the Renaissance simulations.

$M_{\text{vir}} < 4 \times 10^6 M_{\odot}$ will have Pop II burst masses set by Equation (6). Halos with masses $M_{\text{vir}} > 4 \times 10^6 M_{\odot}$ instead use the Renaissance-calibrated Pop II burst masses that scale more steeply with the halo mass and redshift. This threshold reflects the physical transition between shallow and deep potential wells: low-mass halos are more susceptible to SN-driven gas loss, limiting the total stellar mass that can form per burst, whereas higher-mass halos can retain a larger fraction of their gas and sustain more efficient, repeated starbursts. Any halo in the bursty stage will have shorter quiescent periods between bursts as it grows more massive, consistent with the fits presented in R. Hazlett et al. (2025) and with the

expectation that deeper potentials shorten recovery times following SN feedback.

Halos undergoing Pop II star formation will then proceed to the steady star formation stage, which occurs once the halos exceed a critical virial temperature of about 2×10^4 K. In this regime, star formation is well described by a bathtub-type model where gas inflow, star formation, and SN-driven outflows regulate the system. Our calibration to Renaissance favors an efficiency of about 1% of gas converted into stars per freefall time, with some scatter from halo to halo. The steady stage reproduces the smoother star formation seen in more massive halos, while still accounting for the variability

introduced by feedback. More details about the Renaissance-calibrated star formation model can be found in Table 1 and in R. Hazlett et al. (2025).

3.3. H II Bubbles and External Metal Enrichment

To include feedback effects from reionization and external metal enrichment, we implemented H II and metal bubbles into the SAM. To determine the size and growth of the ionized H II bubbles, we use the fast Fourier transform method introduced in E. Visbal et al. (2020). We find the total number density of ionizing photons produced in DM halos that have escaped to the surrounding intergalactic medium for each cell in a 256^3 cubic grid. This is set by the total mass of stars ever formed in each cell, along with the escape fraction of hydrogen-ionizing photons from halos hosting Pop III and metal-enriched stars, $f_{\text{esc,III}}$ and $f_{\text{esc,II}}$, respectively. A cell is considered ionized if it exceeds a threshold dependent on the cosmic mean comoving density of hydrogen atoms. T. Behling et al. (2025) found that SAMs that do not account for self-shielded neutral gas in dense filaments overestimate the fraction of halos in H II regions impacted by reionization by up to an order of magnitude. Therefore, it is likely that our calibrated model overestimates the reionization feedback.

Since AEOS only covers the early stages of reionization at redshifts of $z > 14.6$, the extent of ionized H II bubbles remains fairly localized, only impacting Pop III star formation in a handful of halos. Using escape fractions of ionizing radiation of $f_{\text{esc,II}} = 0.02$ for Pop II stars and $f_{\text{esc,III}} = 0.1$ for Pop III stars, we obtain an ionization fraction of ~ 0.05 at $z = 15$ compared to the AEOS value of ~ 0.08 (K. Brauer et al. 2025b). In our model, if a halo with no prior star formation is located in an ionized region, we increase the critical mass for Pop III formation to

$$M_{\text{crit}} = 1.5 \times 10^8 \left(\frac{1+z}{11} \right)^{-3/2} M_{\odot}, \quad (7)$$

consistent with M. Dijkstra et al. (2004).

We also include the external enrichment of pristine halos by SN wind-driven metal bubbles that result in metal-enriched star formation (E. Visbal et al. 2020). If a halo is in a bubble that exceeds our fiducial critical metallicity of $Z_{\text{crit}} = 10^{-5} Z_{\odot}$, the Pop II star formation will proceed as described in Section 3.2.2. Other works find a critical value near our fiducial choice (e.g., V. Bromm & A. Loeb 2003; K. Omukai et al. 2005; B. D. Smith & S. Sigurdsson 2007; B. D. Smith et al. 2009), but it could be significantly lower ($Z_{\text{crit}} \approx 10^{-6} Z_{\odot}$) if dust cooling is important, rather than just C and O (K. Omukai et al. 2005; A.-K. Jappsen et al. 2009).

4. Results

The final snapshot of the AEOS simulations is at $z = 14.6$, while the DM merger trees used for the SAM extend to $z = 8.3$. This allows the SAM to make predictions that go beyond the end of the AEOS simulation, while maintaining the calibration to the AEOS results. At $z = 14.6$, AEOS contains ~ 200 Pop III-forming halos and fewer than 20 Pop II-forming halos. The scarcity of Pop II sources has important implications for interpreting differences with the SAM, since stochastic effects dominate when the number of star-forming halos is small.

We run 10 independent SAM realizations with the fiducial physics summarized in Table 1 and escape fractions of $f_{\text{esc,III}} = 0.1$ and $f_{\text{esc,II}} = 0.02$. Figure 5 shows the total number of halos hosting Pop III and Pop II stars for the different random seeds, as well as varying physical assumptions. Figure 6 shows the corresponding buildup in total stellar mass. For $z > 17$, the fiducial models closely track the AEOS results for both populations, demonstrating the strong consistency of our calibrated model with AEOS.

At $z < 17$, the SAM begins to underestimate Pop III total stellar mass by a factor of ~ 2 and the number of halos forming Pop III by $\sim 20\%$ by $z = 14.6$ (Figure 5). These differences are likely related to increased local reionization effects in the SAM compared to AEOS, which can delay Pop III formation in highly clustered regions. However, the halos that host Pop III stars are mostly consistent in both the SAM and AEOS, as shown in Figure 7. In contrast, most of the halos hosting Pop II stars in the SAM form in different halos compared to AEOS, reflecting a shortcoming in how t_{delay} is implemented. Because the actual distribution of t_{delay} is not fully captured in AEOS, our model cannot employ a physically motivated calibration for this parameter, leading to discrepancies in which halos transition from Pop III to Pop II formation.

For Pop II, the number of star-forming halos matches well at $z = 14.6$, although the total stellar mass of Pop II significantly diverges for $z > 17$. This offset is likely driven by the fact that most Pop II star formation is concentrated in a handful of halos (< 20 halos). In particular, the delay time function is poorly sampled, and longer delay times in a few AEOS halos over the SAM halos will magnify any discrepancy in star formation between the SAM and AEOS.

Next, we test model variations to examine the sensitivity of the SAM to different physical conditions. Increasing the escape fractions to $f_{\text{esc,III}} = 0.5$ and $f_{\text{esc,II}} = 0.1$ (the HighEsc case) causes significant deviations from AEOS: the ionization fraction approaches ~ 0.17 at $z = 15$, the Pop III total stellar mass is underestimated by a factor of ~ 4 , and the number of Pop III halos is underestimated by a factor of ~ 2.5 at $z < 20$. These discrepancies arise because larger H II regions increase the suppression of Pop III formation, shifting the impact of reionization to earlier times. Despite this, the total Pop II stellar mass and number of Pop II star-forming halos remain roughly consistent with the fiducial runs, suggesting that the earlier impact of reionization is likely efficient at suppressing star formation in low-mass halos.

Excluding both reionization and external metal enrichment (the NoReionExt case) leads to higher Pop III total stellar masses and halo counts between $14.6 < z < 25$ compared to the fiducial model, underscoring the regulating role of feedback processes over cosmic time. Pop III in AEOS falls between the NoReionExt and the fiducial SAM results by $z = 14.6$, likely due to overestimates of reionization feedback. However, by $z = 14.6$ the results converge, once again underscoring the regulating role of feedback processes over cosmic time. In this case, the enhanced Pop III activity at earlier times also boosts Pop II formation for $z > 17$.

Different behavior is seen in the R. Hazlett et al. (2025) realization, calibrated only to the Renaissance simulations. Because the Renaissance volumes cannot resolve the 10^5 – $10^6 M_{\odot}$ minihalos thought to host Pop III star formation (M. Kulkarni et al. 2021; A. T. P. Schauer et al. 2021; O. Nebrin et al. 2023), an increase in M_{crit} near the resolution

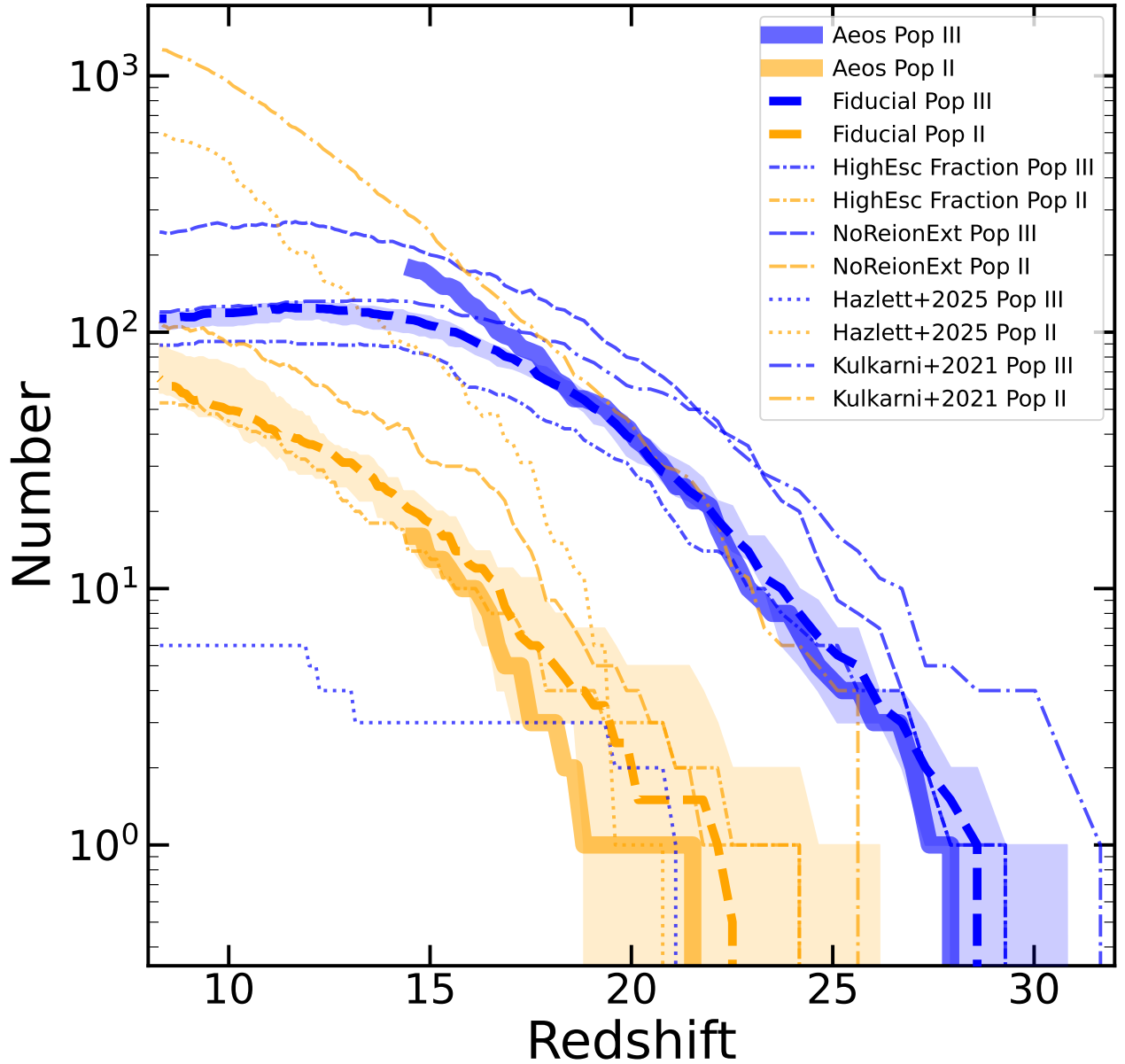


Figure 5. Total number of halos hosting Pop III (blue) and Pop II stars (yellow). The results from AEOS are shown by thick solid lines for Pop III and Pop II, respectively. The results of 10 realizations of our fiducial SAM for Pop III and Pop II are also shown, with the dashed line representing the average and the shaded region extending between the maximum and minimum total stellar masses of the 10 simulations. The HighEsc densely dotted–dashed lines show a realization using higher ionizing radiation escape fractions. The NoReionExt densely dashed lines show the impact of excluding reionization and external enrichment from the SAM. The Hazlett+2025 dotted lines show results from the SAM calibrated only to the Renaissance simulations. The Kulkarni+2021 dotted–dashed lines show the impact of using the unmodified M_{crit} from M. Kulkarni et al. (2021).

limit was introduced in that calibration. This suppresses early Pop III formation, delaying its onset and producing nearly two orders of magnitude fewer Pop III halos than AEOS by $z = 14.6$. Furthermore, the Renaissance-only calibration uses a broader distribution for the Pop III–Pop II delay time t_{delay} , including delays much shorter than the t_{delay} distribution calibrated to AEOS. This results in a more rapid transition to Pop II formation after the first Pop III stars compared to our fiducial SAM parameters.

Finally, the M. Kulkarni et al. (2021) realization illustrates the impact of our modified critical mass threshold M_{crit} (Section 3.2.2). Without this correction, the SAM produces an artificial excess of low-mass Pop III halos, much earlier than the fiducial SAM and AEOS. The use of the unmodified M_{crit} results in three times as many Pop III-forming halos as AEOS

at $z > 17$, which in turn produces more numerous Pop II hosts earlier. For example, at $z = 20$ there are already ~ 70 Pop II halos in the M. Kulkarni et al. (2021) realization of the SAM, versus only 1–2 in AEOS and the fiducial SAM. The total stellar mass predictions show smaller offsets in total Pop III mass, which is consistent with AEOS at $z > 20$ and is about a factor of three lower at $z < 20$, but the total Pop II mass is systematically overestimated.

Overall, our fiducial SAM realizations achieve excellent agreement with AEOS across both stellar populations and redshifts, with only modest deviations attributable to small-scale reionization effects and the limited number of Pop II halos in AEOS. The comparisons emphasize the necessity of our M_{crit} modification, which substantially improves consistency at high redshift and demonstrates that the SAM provides

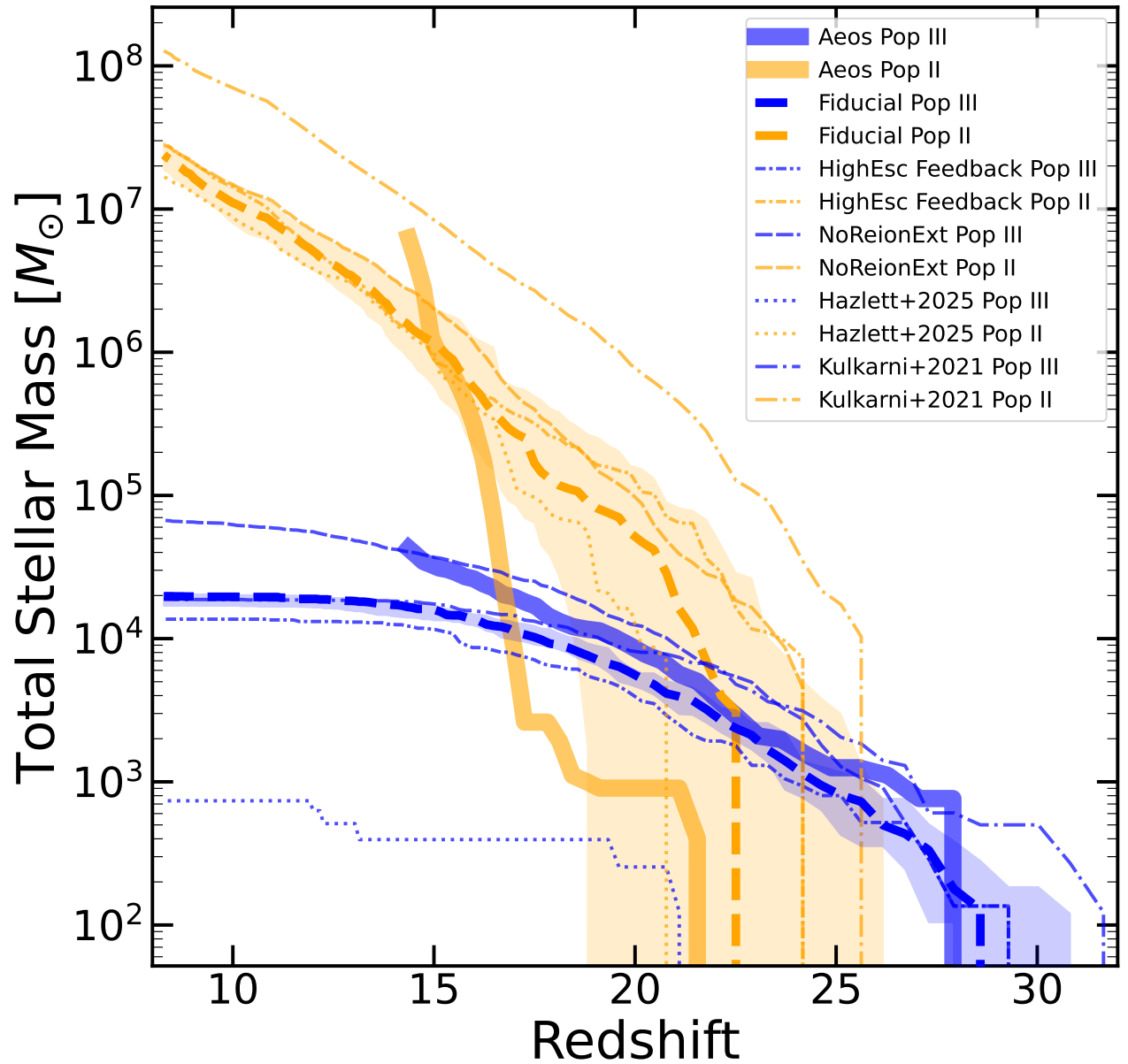


Figure 6. Total stellar masses for Pop III and metal-enriched Pop II stars following the same convention for lines as in the previous figure.

a robust framework for extending predictions beyond the AEOS volume to later cosmic times.

5. Application to Dark-matter-only Simulations

We apply our SAM to 10 cosmological N -body simulations produced using GADGET2 (V. Springel et al. 2001). Each simulation is 3 Mpc on a side and has 512^3 particles corresponding to a particle mass of $8 \times 10^3 M_\odot$. In each simulation, ~ 100 snapshots were saved from $z = 40$ to $z = 6$, spaced in cosmic time by $\Delta t = t_H/40$, where t_H is the Hubble time at the previous snapshot. This time spacing corresponds to ≈ 0.25 of the dynamical time of a halo at its virial radius.

Within each simulation, we identify the number density of active Pop III sources, which we define as halos that have formed stars within the past 3 Myr. To ensure that we remain within the calibration regime of the Renaissance and AEOS simulations, we impose a conservative halo mass ceiling of $10^9 M_\odot$. Once this threshold is reached, the SAM halts. Across

the 10 realizations, our SAM evolves self-consistently down to $z \sim 10$, at which point the upper halo mass limit is reached.

Figure 8 shows the number densities of active Pop III sources with ages less than 3 Myr predicted by our fiducial SAM and the results of E. Visbal et al. (2025) that compared a prior implementation of the SAM with observations of LAP1-B, a Pop III candidate system at $z = 6.6$ that is gravitationally lensed by the galaxy cluster MACS J0416 (E. Vanzella et al. 2023b; K. Nakajima et al. 2026). Both models were applied to the same 10 N -body simulations.

Our updated SAM predicts higher number densities of active Pop III sources than E. Visbal et al. (2025) for $z > 15$. This difference arises because E. Visbal et al. (2025) use M_{crit} from M. Kulkarni et al. (2021) and incorporate a nonzero v_{bc} . This results in delayed Pop III star formation at high redshift. At $z = 10$ our SAM produces a number density of active Pop III sources of $\sim 0.04 \text{ Mpc}^{-3}$. Following from E. Visbal et al. (2025) for LAP1-B, this corresponds to ~ 10 Pop III sources formed within 3 Myr at $z = 10$ that are within the lensed area

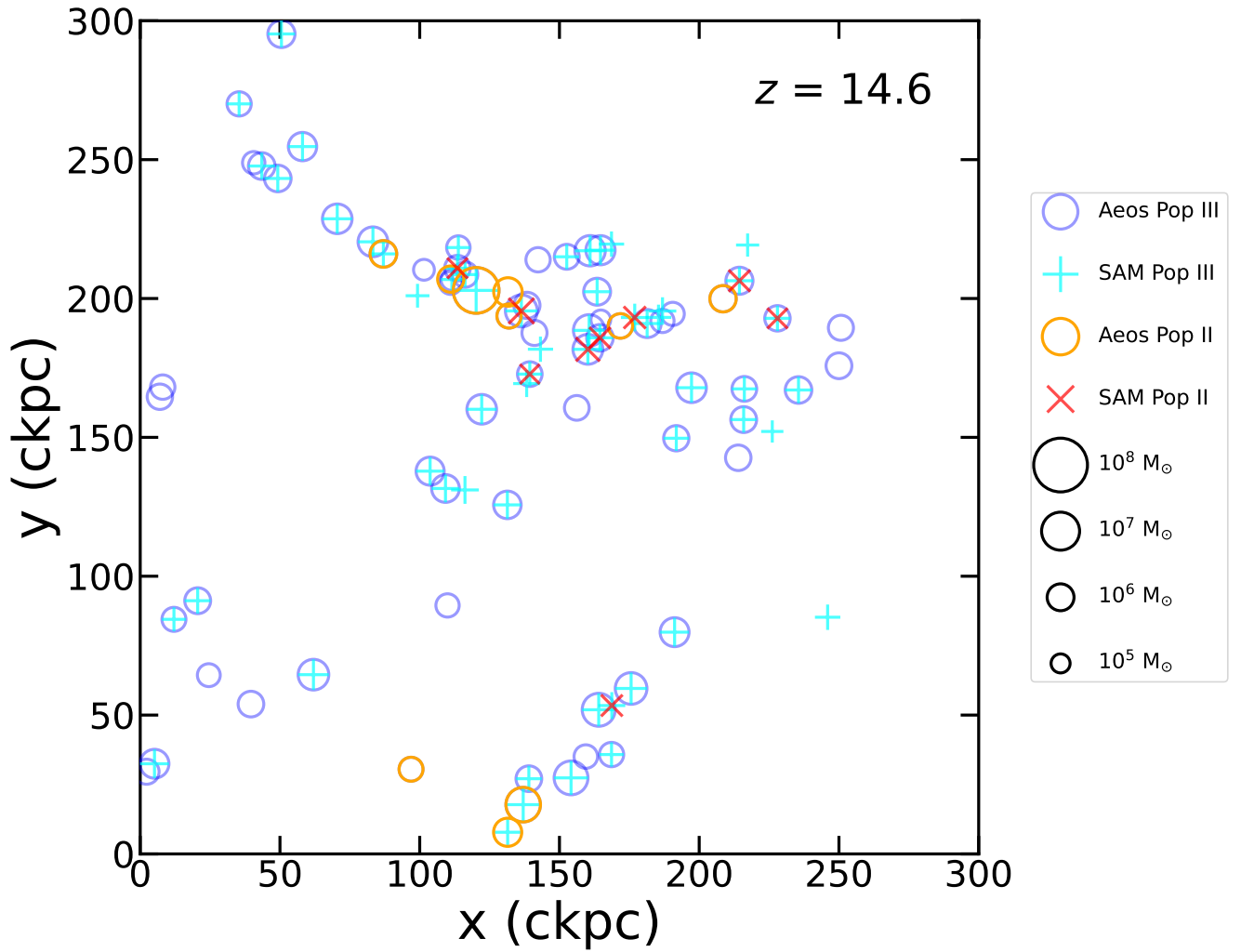


Figure 7. A subregion of the AEOS volume at the final simulation snapshot scaled in comoving kiloparsecs. Halos that have formed Pop III and Pop II stars in AEOS are represented by open blue and orange circles. Halos that have formed Pop III and Pop II stars in the SAM are represented by cyan plus signs and red crosses. The size of the marker is scaled by the halo mass. Overlapping open circles and plus signs or crosses are halos that have formed either Pop III or Pop II in both AEOS and the SAM. Halos that have not formed stars are not shown.

where the foreground galaxy cluster MACS J0416 provides a magnification of $\mu > 30$.

6. Conclusions

In this work, we present an SAM for the formation of the first stars and galaxies, calibrated against two complementary cosmological hydrodynamical simulations. The AEOS simulation resolves individual Pop III stars and their feedback in unprecedented detail, while the Renaissance simulations follow metal-enriched Pop II star formation in a variety of environments to $z \sim 11$. By calibrating to both, our SAM is uniquely positioned to capture the small-scale physics of primordial star formation and span a wide range in halo mass. This dual calibration allows us to build a predictive framework that bridges the gap between local star formation processes and the global history of the earliest galaxies.

A key result of our study is the close agreement between the SAM and AEOS for Pop III star formation. Across 10 realizations of our fiducial model, the SAM accurately reproduces the evolution of total Pop III stellar mass and the number of halos hosting Pop III stars, with particularly strong agreement at $z > 17$. Where small discrepancies arise, for

example, the underestimation of Pop III stellar mass and halo counts at $z < 17$, we identify their likely physical origin in local reionization effects. Importantly, the halos that form Pop III stars in AEOS are predominantly the same halos that do so in the SAM, underscoring the physical fidelity of our approach. The large difference in Pop II stellar mass between the SAM and AEOS for $z < 17$ is likely due to the small number (< 20) of AEOS halos hosting Pop II star formation, which limits the robustness of our calibration to AEOS. A couple of AEOS halos having longer values of the time delay between the death of Pop III stars and the onset of Pop II formation in a given halo t_{delay} than the SAM could produce the difference. Despite the discrepancy in Pop II stellar mass, the number of Pop II-forming halos predicted by the fiducial models agrees well with AEOS.

Another central finding is the necessity of modifying the critical mass threshold M_{crit} used to regulate Pop III star formation if a scatter is applied to this criterion similar to that seen in AEOS. We demonstrate that, without this modification, applying a scatter to M_{crit} in the SAM produces an artificial excess of low-mass halos forming Pop III stars at high redshift because there are more low-mass halos below the cutoff than higher-mass halos above it. By applying our corrected

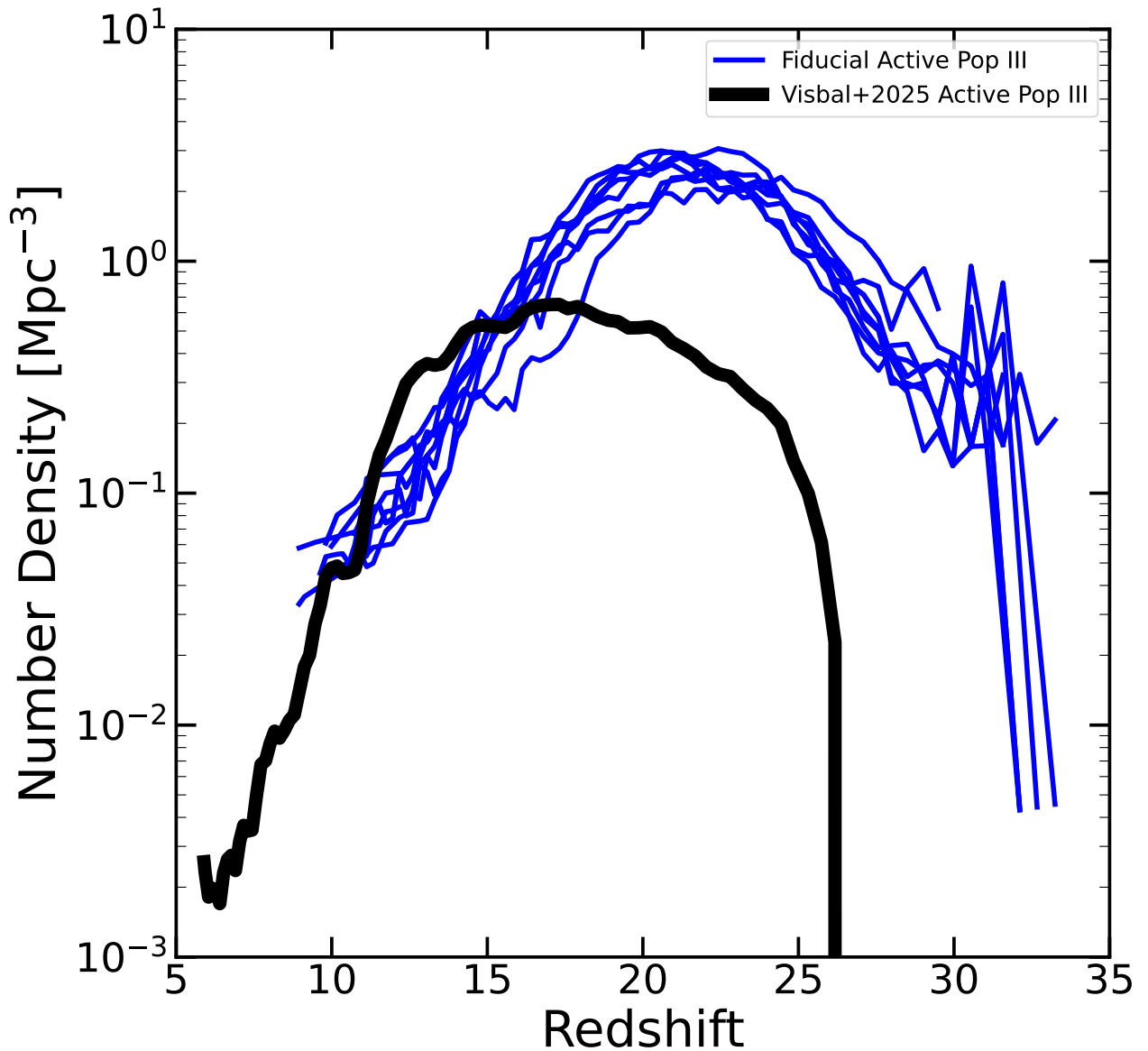


Figure 8. Number density of Pop III sources younger than 3 Myr as a function of redshift for the fiducial SAM realizations run on the 10 N -body simulations, shown by the blue lines. The black line is the number density of Pop III sources younger than 3 Myr from E. Visbal et al. (2025) that was compared to observations of LAP1-B using a prior implementation of the SAM.

prescription, we achieve significantly better agreement with AEOS in both the total stellar mass and the number of star-forming halos. This highlights that the function for M_{crit} presented by M. Kulkarni et al. (2021) cannot be applied directly in SAMs without adjustment. Our modification offers a general solution for future applications.

We also address an important nuance in calibrating t_{delay} . When relying on simulation outputs, it is crucial to account not only for halos where Pop II stars have already formed but also for halos that have hosted Pop III stars and are still awaiting Pop II star formation, either because of intrinsically long t_{delay} values or because the simulation ends before the transition occurs. Our SAM incorporates this by implementing a simple prescription for t_{delay} that successfully captures the expected Pop III–Pop II transition and produces a delay time distribution consistent with the AEOS sample. However, we find that most of the halos hosting Pop II stars in the SAM form in different halos compared to AEOS, reflecting a shortcoming in our implementation of t_{delay} . Because the actual

distribution of t_{delay} is not fully captured in AEOS, our model cannot use a physically motivated calibration for this parameter. The importance of t_{delay} to the overall star formation history motivates dedicated simulations extending on the work of M. Jeon et al. (2014).

Finally, we extend our SAM beyond the resolution and temporal limits of AEOS by applying it to DM-only simulations. This allows us to make predictions for the abundance of halos hosting Pop III stars down to $z \sim 10$, reaching redshifts directly relevant for comparison with JWST observations of candidate Pop III galaxies. In agreement with E. Visbal et al. (2025), at $z = 10$ we find that roughly 10 Pop III sources formed within the past 3 Myr lie within the area lensed by the galaxy cluster MACS J0416, where the magnification exceeds $\mu > 30$. This finding provides additional support for the prospect that both strongly lensed Pop III galaxies (e.g., LAP1-B) and individual Pop III stars may be detectable with JWST. However, our use of a streaming velocity $v_{\text{bc}} = 0 \text{ km s}^{-1}$ to match AEOS is not

representative of the conditions in a typical patch of the Universe of similar size, which at recombination will have $v_{bc} = 30 \text{ km s}^{-1}$ (D. Tselikhovich & C. Hirata 2010).

In summary, we describe an SAM that is (1) calibrated to two complementary hydrodynamical simulations, (2) physically consistent with small-scale primordial star formation, and (3) applicable to large-volume DM-only simulations. Unlike full hydrodynamical runs, our SAM can be evolved efficiently to lower redshifts directly relevant for JWST, allowing predictions to be compared against candidate Pop III galaxies. Its flexibility enables systematic tests of different prescriptions for star formation, reionization, and metal enrichment, and it can be applied to volumes large enough to begin probing the cosmological environment of primordial star formation. Critically, the SAM executes in a small fraction of the time required for full hydrodynamical simulations, running in less than a day on a laptop, while still incorporating the more realistic physics of the hydrodynamical simulations through calibration. This combination of speed, physical fidelity, and scalability makes our SAM a powerful framework for exploring the earliest stages of galaxy formation.

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Facility: TACC.

Software: NUMPY (C. R. Harris et al. 2020), SCIPY (P. Virtanen et al. 2020), MATPLOTLIB (J. D. Hunter 2007), CYTHON (S. Behnel et al. 2011), Jupyter notebook (T. Kluyver et al. 2016), ROCKSTAR (P. S. Behroozi et al. 2013a), CONSISTENT TREES (P. S. Behroozi et al. 2013b), YT (M. J. Turk et al. 2011), YTREE (B. Smith & M. Lang 2019).

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References

- Abel, T., Bryan, G. L., & Norman, M. L. 2002, *Sci*, **295**, 93
 Adamo, A., Bradley, L. D., Vanzella, E., et al. 2024, *Natur*, **632**, 513
 Ahn, K., Shapiro, P. R., Iliev, I. T., Mellema, G., & Pen, U.-L. 2009, *ApJ*, **695**, 1430
 Andersson, E. P., Agertz, O., Renaud, F., & Teyssier, R. 2023, *MNRAS*, **521**, 2196
 Andersson, E. P., Rey, M. P., Pontzen, A., et al. 2025, *ApJ*, **978**, 129
 Behling, T., Hazlett, R., Kulkarni, M., & Visbal, E. 2025, arXiv:2508.04808
 Behnel, S., Bradshaw, R., Citro, C., et al. 2011, *CSE*, **13**, 31
 Behroozi, P. S., Wechsler, R. H., & Wu, H.-Y. 2013a, *ApJ*, **762**, 109
 Behroozi, P. S., Wechsler, R. H., Wu, H.-Y., et al. 2013b, *ApJ*, **763**, 18
 Brauer, K., Emerick, A., Mead, J., et al. 2025a, *ApJ*, **980**, 41
 Brauer, K., Mead, J., Wise, J. H., et al. 2025b, *ApJ*, **993**, 2
 Bressan, A., Marigo, P., Girardi, L., et al. 2012, *MNRAS*, **427**, 127
 Bromm, V., Coppi, P. S., & Larson, R. B. 2002, *ApJ*, **564**, 23
 Bromm, V., & Loeb, A. 2003, *Natur*, **425**, 812
 Bromm, V., & Loeb, A. 2004, *NewA*, **9**, 353
 Brummel-Smith, C., Bryan, G., Butsky, I., et al. 2019, *JOSS*, **4**, 1636
 Bryan, G. L., Norman, M. L., O'Shea, B. W., et al. 2014, *ApJS*, **211**, 19
 Cai, S., Li, M., Cai, Z., et al. 2025, *ApJL*, **993**, L52
 Clark, P. C., Glover, S. C. O., Smith, R. J., et al. 2011, *Sci*, **331**, 1040
 Corazza, L. C., Miranda, O. D., & Wuensche, C. A. 2022, *A&A*, **668**, A191
 Crosby, B. D., O'Shea, B. W., Smith, B. D., Turk, M. J., & Hahn, O. 2013, *ApJ*, **773**, 108
 de Bannassuti, M., Salvadori, S., Schneider, R., Valiante, R., & Omukai, K. 2017, *MNRAS*, **465**, 926
 Dekel, A., & Rees, M. J. 1987, *Natur*, **326**, 455
 Dijkstra, M., Haiman, Z., Rees, M. J., & Weinberg, D. H. 2004, *ApJ*, **601**, 666
 Emerick, A., Bryan, G. L., & Mac Low, M.-M. 2019, *MNRAS*, **482**, 1304
 Ezzeddine, R., Frebel, A., Roederer, I. U., et al. 2019, *ApJ*, **876**, 97
 Feathers, C. R., Kulkarni, M., Visbal, E., & Hazlett, R. 2024, *ApJ*, **962**, 62
 Fernandez, R., Bryan, G. L., Haiman, Z., & Li, M. 2014, *MNRAS*, **439**, 3798
 Frebel, A., & Norris, J. E. 2015, *ARA&A*, **53**, 631
 Fujimoto, S., Naidu, R. P., Chisholm, J., et al. 2025, *ApJ*, **989**, 46
 Geha, M., Brown, T. M., Tumlinson, J., et al. 2013, *ApJ*, **771**, 29
 Gessey-Jones, T., Sartorio, N. S., Fialkov, A., et al. 2022, *MNRAS*, **516**, 841
 Gnedin, N. Y. 2000, *ApJ*, **542**, 535
 Gnedin, N. Y., & Hui, L. 1998, *MNRAS*, **296**, 44
 Greif, T. H., Bromm, V., Clark, P. C., et al. 2012, *MNRAS*, **424**, 399
 Greif, T. H., Springel, V., White, S. D. M., et al. 2011, *ApJ*, **737**, 75
 Gutcke, T. A., Pakmor, R., Naab, T., & Springel, V. 2021, *MNRAS*, **501**, 5597
 Haardt, F., & Madau, P. 2012, *ApJ*, **746**, 125
 Haiman, Z., Abel, T., & Rees, M. J. 2000, *ApJ*, **534**, 11
 Haiman, Z., Rees, M. J., & Loeb, A. 1996, *ApJ*, **467**, 522
 Haiman, Z., Rees, M. J., & Loeb, A. 1997, *ApJ*, **476**, 458
 Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Natur*, **585**, 357
 Hartwig, T., Magg, M., Chen, L.-H., et al. 2022, *ApJ*, **936**, 45
 Hartwig, T., Yoshida, N., Magg, M., et al. 2018, *MNRAS*, **478**, 1795
 Hazlett, R., Kulkarni, M., Visbal, E., & Wise, J. H. 2025, *ApJ*, **978**, 13
 Hegde, S., & Furlanetto, S. R. 2023, *MNRAS*, **525**, 428
 Heger, A., & Woosley, S. E. 2010, *ApJ*, **724**, 341
 Hirano, S., Hosokawa, T., Yoshida, N., et al. 2014, *ApJ*, **781**, 60
 Hirano, S., Hosokawa, T., Yoshida, N., Omukai, K., & Yorke, H. W. 2015, *MNRAS*, **448**, 568
 Hoeft, M., Yepes, G., Gottlöber, S., & Springel, V. 2006, *MNRAS*, **371**, 401
 Hosokawa, T., Hirano, S., Kuiper, R., et al. 2016, *ApJ*, **824**, 119
 Hosokawa, T., Omukai, K., Yoshida, N., & Yorke, H. W. 2011, *Sci*, **334**, 1250
 Hunter, J. D. 2007, *CSE*, **9**, 90
 Inayoshi, K., Kashiyama, K., Visbal, E., & Haiman, Z. 2016, *MNRAS*, **461**, 2722
 Ishiyama, T., & Hirano, S. 2025, *ApJ*, **994**, 107
 Jappsen, A.-K., Klessen, R. S., Glover, S. C. O., & Mac Low, M.-M. 2009, *ApJ*, **696**, 1065
 Jaura, O., Glover, S. C. O., Wollenberg, K. M. J., et al. 2022, *MNRAS*, **512**, 116
 Jeon, M., & Ko, M. 2026, *MNRAS*, **545**, staf2158
 Jeon, M., Pawlik, A. H., Bromm, V., & Milosavljević, M. 2014, *MNRAS*, **444**, 3288
 Ji, A. P., Curtis, S., Storm, N., et al. 2024, *ApJL*, **961**, L41
 Klessen, R. S., & Glover, S. C. O. 2023, *ARA&A*, **61**, 65
 Kluyver, T., Ragan-Kelley, B., Pérez, F., et al. 2016, in *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, ed. F. Loizides & B. Schmidt (IOS Press), 87

- Komiya, Y., Habe, A., Suda, T., & Fujimoto, M. Y. 2010, *ApJ*, 717, 542
- Kroupa, P. 2001, *MNRAS*, 322, 231
- Kulkarni, M., Visbal, E., & Bryan, G. L. 2021, *ApJ*, 917, 40
- Lahén, N., Naab, T., Johansson, P. H., et al. 2020, *ApJ*, 891, 2
- Lahén, N., Naab, T., Kauffmann, G., et al. 2023, *MNRAS*, 522, 3092
- Lake, W., Grudić, M. Y., Naoz, S., et al. 2025, *ApJL*, 985, L6
- Lanz, T., & Hubeny, I. 2003, *ApJS*, 146, 417
- Latif, M. A., Schleicher, D. R. G., Schmidt, W., & Niemeyer, J. 2013, *ApJL*, 772, L3
- Latif, M. A., Whalen, D., & Khochfar, S. 2022, *ApJ*, 925, 28
- Liu, B., & Bromm, V. 2020, *MNRAS*, 497, 2839
- Liu, B., Hartwig, T., Sartorio, N. S., et al. 2024, *MNRAS*, 534, 1634
- Machacek, M. E., Bryan, G. L., & Abel, T. 2001, *ApJ*, 548, 509
- Maiolino, R., Übler, H., Perna, M., et al. 2024, *A&A*, 687, A67
- Mead, J., Brauer, K., Bryan, G. L., et al. 2025a, *ApJ*, 980, 62
- Mead, J., Brauer, K., Bryan, G. L., et al. 2025b, arXiv:2509.13580
- Mirocha, J., Mebane, R. H., Furlanetto, S. R., Singal, K., & Trinh, D. 2018, *MNRAS*, 478, 5591
- Morishita, T., Liu, Z., Stiavelli, M., et al. 2025, arXiv:2507.10521
- Moriya, T. J., Wong, K. C., Koyama, Y., et al. 2019, *PASJ*, 71, 59
- Mowla, L., Iyer, K., Asada, Y., et al. 2024, *Natur*, 636, 332
- Nakajima, K., Ouchi, M., Harikane, Y., et al. 2026, *Natur*, 653, 363
- Nebrin, O., Giri, S. K., & Mellema, G. 2023, *MNRAS*, 524, 2290
- Noh, Y., & McQuinn, M. 2014, *MNRAS*, 444, 503
- Okamoto, T., Gao, L., & Theuns, T. 2008, *MNRAS*, 390, 920
- Omukai, K., Tsuribe, T., Schneider, R., & Ferrara, A. 2005, *ApJ*, 626, 627
- O'Shea, B. W., & Norman, M. L. 2008, *ApJ*, 673, 14
- O'Shea, B. W., Wise, J. H., Xu, H., & Norman, M. L. 2015, *ApJL*, 807, L12
- Parsons, J., Mas-Ribas, L., Sun, G., et al. 2022, *ApJ*, 933, 141
- Planck Collaboration, Ade, P. A. R., Aghanim, N., et al. 2014, *A&A*, 571, A16
- Qin, Y., Mesinger, A., Park, J., Greig, B., & Muñoz, J. B. 2020, *MNRAS*, 495, 123
- Safarzadeh, M., & Haiman, Z. 2020, *ApJL*, 903, L21
- Salpeter, E. E. 1955, *ApJ*, 121, 161
- Schaerer, D. 2002, *A&A*, 382, 28
- Schauer, A. T. P., Drory, N., & Bromm, V. 2020, *ApJ*, 904, 145
- Schauer, A. T. P., Glover, S. C. O., Klessen, R. S., & Clark, P. 2021, *MNRAS*, 507, 1775
- Shapiro, P. R., Giroux, M. L., & Babul, A. 1994, *ApJ*, 427, 25
- Sharda, P., Federrath, C., & Krumholz, M. R. 2020, *MNRAS*, 497, 336
- Sharda, P., & Menon, S. H. 2025, *MNRAS*, 540, 1745
- Skúladóttir, Á., Koutsouridou, I., Vanni, I., et al. 2024, *ApJL*, 968, L23
- Skúladóttir, Á., Salvadori, S., Amarsi, A. M., et al. 2021, *ApJL*, 915, L30
- Smith, B., & Lang, M. 2019, *JOSS*, 4, 1881
- Smith, B. D., Bryan, G. L., Glover, S. C. O., et al. 2017, *MNRAS*, 466, 2217
- Smith, B. D., & Sigurdsson, S. 2007, *ApJL*, 661, L5
- Smith, B. D., Turk, M. J., Sigurdsson, S., O'Shea, B. W., & Norman, M. L. 2009, *ApJ*, 691, 441
- Smith, M. C. 2021, *MNRAS*, 502, 5417
- Sobacchi, E., & Mesinger, A. 2013, *MNRAS*, 432, L51
- Sodini, A., D'Odorico, V., Salvadori, S., et al. 2024, *A&A*, 687, A314
- Springel, V., Yoshida, N., & White, S. D. M. 2001, *NewA*, 6, 79
- Stacy, A., Greif, T. H., & Bromm, V. 2010, *MNRAS*, 403, 45
- Susa, H., Hasegawa, K., & Tominaga, N. 2014, *ApJ*, 792, 32
- Tegmark, M., Silk, J., Rees, M. J., et al. 1997, *ApJ*, 474, 1
- Thoul, A. A., & Weinberg, D. H. 1996, *ApJ*, 465, 608
- Toma, K., Yoon, S.-C., & Bromm, V. 2016, *SSRv*, 202, 159
- Trenti, M., & Stiavelli, M. 2009, *ApJ*, 694, 879
- Tselikhovich, D., & Hirata, C. 2010, *PhRvD*, 82, 083520
- Turk, M. J., Smith, B. D., Oishi, J. S., et al. 2011, *ApJS*, 192, 9
- Vanzella, E., Claeysens, A., Welch, B., et al. 2023a, *ApJ*, 945, 53
- Vanzella, E., Loiacono, F., Bergamini, P., et al. 2023b, *A&A*, 678, A173
- Ventura, E. M., Qin, Y., Balu, S., & Wyithe, J. S. B. 2024, *MNRAS*, 529, 628
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *NatMe*, 17, 261
- Visbal, E., Bryan, G. L., & Haiman, Z. 2020, *ApJ*, 897, 95
- Visbal, E., Bryan, G. L., & Haiman, Z. 2026, *JCAP*, 02, 077
- Visbal, E., Haiman, Z., & Bryan, G. L. 2015a, *MNRAS*, 450, 2506
- Visbal, E., Haiman, Z., & Bryan, G. L. 2015b, *MNRAS*, 453, 4456
- Visbal, E., Haiman, Z., Terrazas, B., Bryan, G. L., & Barkana, R. 2014, *MNRAS*, 445, 107
- Visbal, E., Hazlett, R., & Bryan, G. L. 2025, *ApJL*, 993, L17
- Wang, X., Cheng, C., Ge, J., et al. 2024, *ApJL*, 967, L42
- Windhorst, R. A., Timmes, F. X., Wyithe, J. S. B., et al. 2018, *ApJS*, 234, 41
- Wise, J. H., & Abel, T. 2007, *ApJ*, 671, 1559
- Wise, J. H., Turk, M. J., Norman, M. L., & Abel, T. 2012, *ApJ*, 745, 50
- Xing, Q.-F., Zhao, G., Liu, Z.-W., et al. 2023, *Natur*, 618, 712
- Yong, D., Kobayashi, C., Da Costa, G. S., et al. 2021, *Natur*, 595, 223
- Yoshida, N., Omukai, K., & Hernquist, L. 2008, *Sci*, 321, 669